MACHINE LEARNING PROJECT REPORT

Binary Person Segmentation Using U-Net and Controlled Data Augmentation

A Controlled Multi-Seed Study Comparing Unaugmented Baseline, Geometric Transformations (Horizontal Flip & Small Zoom), and Instance-Level Copy-Paste Augmentation on the Penn-Fudan Pedestrian Dataset

Student Name	[Student Name]
Roll Number	[Roll Number]
Course / Program	[Course Name & Code]
Department	[Department Name]
Institution	[Institution Name]
Project Supervisor	[Guide / Instructor Name]
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Abstract

Pixel-level person segmentation is a core computer vision capability for pedestrian monitoring, autonomous driving, and intelligent surveillance. However, training deep encoder-decoder networks on small, real-world pedestrian datasets often suffers from severe overfitting and high sensitivity to random initialization. This project investigates binary person segmentation on the Penn-Fudan Pedestrian dataset (170 street photographs containing 345 annotated person instances) using a compact 5-level U-Net (7,763,041 parameters) trained with an equal-weight Binary Cross-Entropy plus soft Dice loss (`DiceBCELoss`).

To isolate the true impact of training-data diversity without introducing architectural complexity or test-set leakage, all models share an identical U-Net backbone, a fixed stratified 119/25/26 (train/validation/test) split, and a strict checkpoint selection protocol based solely on validation Dice. Initial 30-epoch CPU experiments revealed that naive zoom-out augmentation (0.9–1.1×) padded black borders, double-resampling blurred edges, and constant learning rates caused under-convergence. Addressing these bottlenecks on an RTX 4060 GPU with single-step raw stretch resizing, zoom-in scaling (1.0–1.15× at $p = 0.5$), 50 training epochs, and cosine annealing yielded a decisive improvement across three independent training seeds (42, 43, 44).

Compared to the unaugmented baseline (`none`: validation Dice 0.8078 ± 0.0062, IoU 0.6776 ± 0.0087; held-out test Dice 0.8495 ± 0.0130, IoU 0.7385 ± 0.0197), combining horizontal flip and small zoom (`both`) raised validation Dice to 0.8298 ± 0.0087 (+2.20 pp) and IoU to 0.7091 ± 0.0127 (+3.15 pp), while lifting held-out test Dice to 0.8668 ± 0.0016 (+1.73 pp) and IoU to 0.7649 ± 0.0025 (+2.64 pp, paired permutation $p = 0.00130$ across 26 test images) and reducing test IoU seed standard deviation nearly eight-fold. Extending the pipeline with training-only person-instance Copy-Paste augmentation (290 cutouts extracted strictly from the 119 training images) further improved performance to 0.8514 ± 0.0091 validation Dice (0.7414 IoU) and 0.8726 ± 0.0045 test Dice (0.7740 IoU, 22/26 test images improved, $p = 0.00049$).

Keywords: Binary Person Segmentation, U-Net, Penn-Fudan Pedestrian Dataset, Data Augmentation, Copy-Paste Augmentation, Dice Loss, Intersection over Union.

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CHAPTER 1

1. Introduction

1.1 Background and Motivation

Semantic image segmentation assigns a class label to every pixel in a photograph, producing dense spatial masks that delineate exact object silhouettes rather than coarse bounding boxes. Person segmentation—separating human figures from complex street, indoor, and urban backgrounds—plays a foundational role in pedestrian safety monitoring, autonomous navigation, smart surveillance, crowd analytics, and human-computer interaction.

Although modern deep encoder-decoder networks such as U-Net excel at pixel-wise localization, their generalization degrades sharply when labeled training data are scarce. Collecting and pixel-annotating pedestrian datasets is expensive and time-consuming: the widely used Penn-Fudan Pedestrian dataset contains only 170 images (119 training images under a standard 70/15/15 split). Without sufficient visual diversity, a neural network memorizes background textures, lighting patterns, and fixed pedestrian poses in the training split, causing validation loss to plateau while training loss approaches zero. Data augmentation offers a zero-annotation-cost mechanism to synthesize plausible variations in orientation, scale, and scene composition directly during training.

1.2 Problem Statement

Given an RGB street photograph $X \in \mathbb{R}^{3\times H\times W}$, the objective of binary person segmentation is to predict a pixel-level probability map $P \in [0, 1]^{H\times W}$ and thresholded binary mask $M \in \{0, 1\}^{H\times W}$ where all human pixels (including overlapping or occluded pedestrians) belong to the single foreground class (1) and all non-person pixels belong to background (0). Specifically, this project investigates whether controlled data augmentation—starting with horizontal flipping and small zoom, and extending to training-only person-instance Copy-Paste—measurably improves U-Net generalization, boundary accuracy, and seed-to-seed stability on a small pedestrian dataset under identical architecture, loss, and split constraints.

1.3 Objectives

- Audit the 170-image Penn-Fudan Pedestrian dataset for file integrity, instance IDs, and class imbalance, and construct a reproducible, leak-free 70/15/15 (119/25/26) split.
- Evaluate preprocessing design choices empirically, specifically comparing stretched 256 × 256 resizing against aspect-preserving letterbox padding.
- Implement a compact 5-level U-Net baseline with combined Binary Cross-Entropy and soft Dice loss without augmentation.
- Design and ablate geometric augmentations (`zoom` only, `flip` only, and `both`) plus instance-level Copy-Paste augmentation (`copypaste`, `both + copypaste`) across multiple random seeds (42, 43, 44).
- Quantify model gains using pooled Precision, Recall, Dice/F1, and IoU, validate improvements with paired per-image statistical tests, and interpret failure modes on held-out test masks.

1.4 Scope of the Work

This study focuses strictly on **binary semantic person segmentation** (all people merged into one foreground mask) rather than multi-instance tracking or pose estimation. All experiments use a fixed 256 × 256 network input resolution and a 16-base-channel U-Net trained from scratch on Penn-Fudan Pedestrian without external pretraining or extra datasets, ensuring that every observed metric change is attributable solely to training-data augmentation and optimization hygiene.

1.5 Contributions

1. **Leak-Free Dataset Audit and Preprocessing Benchmark:** Audited all 170 image-mask pairs with zero mismatches, established a fixed 119/25/26 split (`split_seed42.json`), and proved experimentally that 256 × 256 stretch resizing outperforms letterbox padding (+0.91 pp validation Dice, +1.32 pp IoU).
2. **Root-Cause Diagnosis of Naive Augmentation Failure:** Identified why Phase 1 CPU zoom-out (0.9–1.1×) lagged baseline—black border padding, double bilinear resampling, and 30-epoch under-convergence—and resolved all three via zoom-in scaling (1.0–1.15×), single-step raw resizing, and 50-epoch cosine learning-rate annealing on GPU.

3. **Controlled 6-Way Multi-Seed Ablation:** Executed 18 GPU training runs (6 augmentation modes × 3 seeds) under identical architecture and loss, demonstrating that `flip + zoom` cuts test IoU seed standard deviation nearly 8× (±0.0197 → ±0.0025).
4. **Training-Only Copy-Paste Extension:** Built an additive, leak-free instance Copy-Paste pipeline (290 cutouts from training images only), achieving the highest validation Dice (0.8514) and held-out test Dice/IoU (0.8726 / 0.7740) while recovering foreground Precision.
5. **Statistical and Qualitative Error Analysis:** Verified significance via paired permutation tests on the 26 held-out test images (p = 0.00049 for IoU) and identified concrete dataset-annotation artifacts (e.g., an unannotated cyclist in `FudanPed00029.png`).

CHAPTER 2

2. Literature Review

2.1 Existing Approaches

Pixel-level image segmentation has evolved from hand-crafted classical heuristics to deep convolutional encoder-decoder networks:

- **Classical Thresholding, Edge, and Region Methods:** Early segmentation algorithms partitioned images using intensity thresholding, Sobel/Canny edge detectors, or watershed region growing. While computationally lightweight, these techniques depend on brittle manual thresholds and fail under outdoor illumination changes, cluttered urban backgrounds, and varied pedestrian clothing.
- **Fully Convolutional Networks (FCN):** Long et al. [1] replaced fully connected classification heads with 1×1 convolutions and transposed upsampling, enabling end-to-end dense prediction. However, deep downsampling discards fine spatial localization, producing coarse object boundaries unless multi-layer skips are fused.
- **U-Net Encoder-Decoder Architecture:** Ronneberger et al. [2] introduced a symmetric U-shaped encoder-decoder with direct skip connections concatenating high-resolution encoder feature maps into corresponding decoder stages. Crucially, Ronneberger et al. demonstrated that strong data augmentation allows U-Net to generalize accurately even on very small annotated datasets.
- **Multi-Scale Context Networks (DeepLabv3+):** Chen et al. [3] combined Atrous Spatial Pyramid Pooling (ASPP) with an encoder-decoder refinement module to capture multi-scale context. While powerful on large benchmarks, DeepLabv3+ introduces substantially higher architectural complexity than a compact U-Net, making it less suitable for isolating pure data-augmentation effects on 119 training images.
- **Loss Functions for Imbalanced Segmentation:** Milletari et al. [4] popularized soft Dice loss to optimize overlap directly under foreground-background class imbalance. Combining Binary Cross-Entropy (smooth pixel gradients) with soft Dice loss (region overlap) is now standard practice for binary segmentation.
- **Data Augmentation and Instance Copy-Paste:** Shorten and Khoshgoftaar [5] surveyed geometric and photometric transformations as primary regularizers against overfitting. For pedestrian and instance-rich scenes, Ghiasi et al. [6] showed that simple Copy-Paste—extracting object instances and pasting them onto other training scenes—provides strong compositional diversity without altering model architecture.
- **Multi-Seed Variance in Semantic Segmentation:** Romera et al. [7] reported typical +1 to +2.5 pp IoU gains from horizontal flipping and multi-scale jitter, while Åkesson et al. [8] emphasized that single-seed segmentation comparisons can be misleading and recommended multi-seed evaluation to separate real improvements from initialization noise.

2.2 Research Gaps

Reviewing the literature alongside the Penn-Fudan Pedestrian benchmark [9] highlights three practical gaps that motivate this project:

1. **Lack of Controlled Augmentation Isolation on Small Datasets:** Many published studies simultaneously change backbone depth, pretraining, input resolution, and augmentation recipes, making it impossible to attribute gains cleanly to individual transformations.
2. **Under-Reported Failure Modes of Naive Scaling:** Standard library defaults (such as zoom-out padding with zero-value black borders or applying transforms to already downsampled images) can silently degrade boundary recall on small datasets unless diagnosed and corrected.
3. **Single-Seed Reporting Bias:** Small-dataset evaluations frequently report only a single lucky random seed, obscuring run-to-run instability. Evaluating every variant across multiple seeds (42, 43, 44) and running paired per-image significance tests addresses this gap directly.

CHAPTER 3

3. Proposed Methodology

3.1 Dataset Description

All experiments use the **Penn-Fudan Pedestrian Dataset** [9], comprising 170 outdoor urban street photographs paired with 170 pixel-level PNG annotation masks. Raw masks encode background as `0` and distinct pedestrian instances as integer IDs `1..8`. For binary person segmentation, all positive instance IDs (`mask > 0`) are merged into a single foreground class (`1`), treating overlapping people as a unified foreground region.

Table 3.1: Penn-Fudan dataset audit summary and fixed 70/15/15 split statistics.

Audit / Split Property	Observed Value	Notes / Verification
Total Image–Mask Pairs	170 pairs	0 missing masks, 0 orphan masks, 0 dimension mismatches
Original Image Dimensions	Width: 253–1017 px; Height: 311–581 px	170 distinct original sizes across the dataset
Person Instances	345 total (1–8 per image)	58 images have 1 person; 1 image has 8 people
Foreground Pixel Fraction	0.1728 ± 0.0766 (~17.3%)	~82.7% background pixels (class-imbalanced)
RGB Channel Statistics ([0, 1])	Mean: [0.4397, 0.4445, 0.4062]	Std Dev: [0.2635, 0.2507, 0.2577]
Fixed Split (`split_seed42.json`)	119 Train / 25 Val / 26 Test	70% / 15% / 15%; 0 filename overlaps across splits

3.2 Data Preprocessing

Every RGB image is scaled to floating-point `[0, 1]` and resized to `256 × 256` using bilinear interpolation, while binary masks are resized using nearest-neighbor interpolation to preserve exact `{0, 1}` labels. To verify whether aspect-ratio distortion harms pedestrian shapes, we compared direct **stretch resizing** against **aspect-preserving letterbox resizing** (zero-padded borders) on validation using seed 42 with horizontal flip (Table 3.2). Because stretch resizing achieved +0.91 pp higher validation Dice and +1.32 pp higher IoU (letterbox inflated Pixel Accuracy artificially via black padding pixels), stretch resizing was selected for all subsequent experiments. Furthermore, in Phase 2–3, online augmentations read directly from raw `data/PennFudanPed/` and apply a **single-step** resize to `256 × 256` to eliminate double-resampling blur.

Table 3.2: Validation comparison of stretch vs. aspect-preserving letterbox resizing (seed 42, flip, CPU, 30 epochs).

Resize Mode	Val Precision	Val Recall	Val Dice / F1	Val IoU	Pixel Accuracy
Stretch (256 × 256)	**0.8312**	**0.8245**	**0.8278**	**0.7062**	**0.9410**
Letterbox (256 × 256 padded)	0.8484	0.7911	0.8187	0.6930	0.9467

3.3 Exploratory Data Analysis (EDA)

Visual inspection and quantitative auditing (Figures 3.1 and 3.2) revealed three important dataset properties: (1) scenes feature diverse outdoor lighting, shadows, bicycles, trees, and brick/glass storefronts; (2) instance counts vary from single pedestrians to dense groups of up to eight overlapping people; and (3) foreground pixels account for only ~17.3% of each image on average, meaning naive Pixel Accuracy (~94%) overstates quality unless paired with Dice/F1 and IoU.



Figure 3.1: Sample Penn-Fudan RGB images and merged binary ground-truth masks.

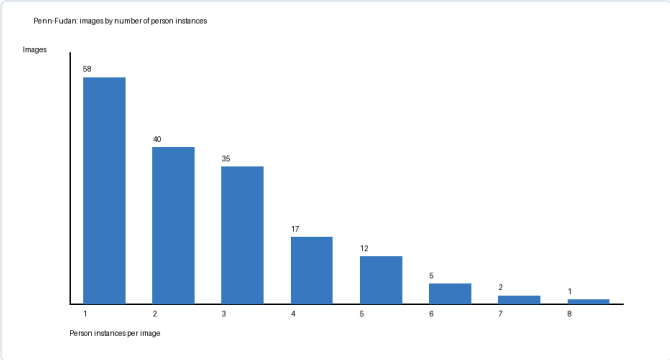


Figure 3.2: Distribution of annotated pedestrian instances per image (170 images).

3.4 Proposed Model Architecture and Loss Function

We employ a compact 5-level **U-Net** (`base_channels = 16`, 7,763,041 trainable parameters) illustrated in Figure 3.3. Each encoder stage applies two `3 × 3` convolutions with Batch Normalization and ReLU activation, followed by `2 × 2` max-pooling across channel widths `16 → 32 → 64 → 128` down to a `256`-channel bottleneck at `16 × 16` spatial resolution. The decoder upsamples via `2 × 2` transposed convolutions, concatenates the matching encoder skip tensor, and refines features through double-convolution blocks back to a `1 × 1` logit map.

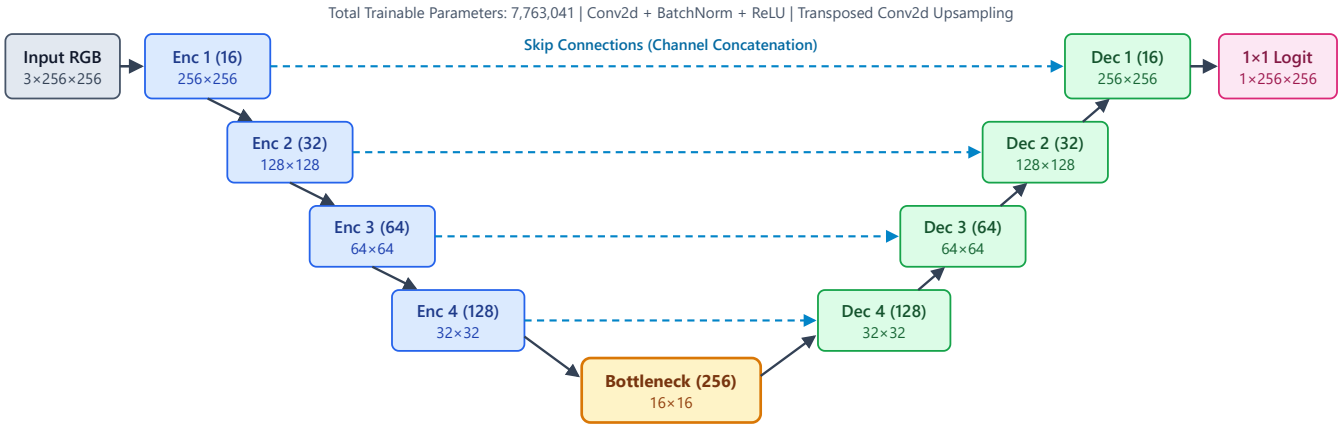


Figure 3.3: 5-level U-Net architecture (`base_channels = 16`, 7,763,041 parameters) with channel-concatenated skip connections.

To balance pixel-wise probability calibration with region-level overlap under class imbalance, training minimizes an equal-weight `DiceBCELoss` (Eqs. 3.1–3.3) with smoothing constant $\varepsilon = 1.0$:

$$L_{BCE} = -\frac{1}{N}\sum_{j=1..N}\left[\, y_j \log(p_j) + (1 - y_j)\log(1 - p_j)\,\right] \tag{3.1}$$

$$L_{Dice} = 1 - \frac{1}{B}\sum_{i=1..B}\frac{2\sum_j p_{ij}\, y_{ij} + \varepsilon}{\sum_j p_{ij} + \sum_j y_{ij} + \varepsilon} \tag{3.2}$$

$$L_{total} = 0.5\, L_{BCE} + 0.5\, L_{Dice} \tag{3.3}$$

3.5 Training and Augmentation Algorithm

Algorithm 3.1: Controlled Multi-Seed U-Net Training with Geometric & Copy-Paste Augmentation

1. **Input:** Fixed split `split_seed42.json` (119 train, 25 val, 26 test), augmentation mode ∈ { `none`, `zoom`, `flip`, `both`, `copypaste`, `both+copypaste` }, seed ∈ {42, 43, 44}.
2. **Pre-extraction (Copy-Paste modes only):** Extract 290 pedestrian cutouts (area ≥ 150 px, $h \geq 20$, $w \geq 8$) strictly from the 119 training images. Validation and test splits remain untouched.
3. **For epoch = 1 to 50:** For each training image-mask pair: (a) If Copy-Paste enabled ($p = 0.5$), paste 1–2 scaled (0.85–1.15×), optionally flipped cutouts at grounded y-positions (overlap ≤ 0.45, 0.8 px feathered RGB edge, crisp binary union mask); (b) If Flip enabled ($p = 0.5$), horizontally flip image and mask; (c) If Zoom enabled ($p = 0.5$), crop a random `1/s` sub-window ($s \in [1.0, 1.15]$) and resize in one step to 256 × 256.
4. Update U-Net weights with Adam (`lr = 0.001`) on `DiceBCELoss`; step `CosineAnnealingLR` ($\eta_{\min} = 10^{-5}$).
5. Evaluate unaugmented validation split; save `best.pt` whenever validation micro Dice improves.
6. **Output:** Best validation checkpoint evaluated once on the held-out 26-image test split.

3.6 Experimental Setup

Table 3.3 summarizes the controlled experimental protocol used across all Phase 2 and Phase 3 runs on an NVIDIA GeForce RTX 4060 Laptop GPU (`torch 2.14.1+cu130`). Every variant was trained identically across seeds 42, 43, and 44 (18 runs total).

Table 3.3: Summary of controlled training configurations and augmentation variants.

Parameter / Variant	Setting
Hardware & Framework	NVIDIA RTX 4060 Laptop GPU, PyTorch 2.14.1+cu130, RAM image caching
Optimization	Adam (`lr = 0.001`), 50 epochs, batch size 4, CosineAnnealingLR (`eta_min = 1e-5`)
Augmentation Modes	`none`, `zoom` (1.0–1.15×, p=0.5), `flip` (p=0.5), `both`, `copypaste` (p=0.5), `both+copypaste`
Evaluation Protocol	Best-validation-Dice checkpoint selection; threshold 0.5; 3-seed Mean ± Sample SD

CHAPTER 4

4. Results and Discussion

4.1 Performance Evaluation Metrics

Model predictions are thresholded at $p \geq 0.5$ and pooled across all pixels in the evaluation split (micro-averaging) to compute True Positives (TP), False Positives (FP), False Negatives (FN), and True Negatives (TN) via Eqs. (4.1)–(4.6):

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{4.1–4.2}$$

$$\text{Dice / F1} = \frac{2\,\text{TP}}{2\,\text{TP} + \text{FP} + \text{FN}}, \quad \text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{4.3–4.4}$$

$$\text{Pixel Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}, \quad \text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \tag{4.5–4.6}$$

4.2 Experimental Results

Initial Phase 1 CPU runs (30 epochs, constant `lr = 0.001`) showed that horizontal flip improved over baseline, but naive 0.9–1.1× zoom at $p = 1.0$ collapsed validation Recall to 0.8022 and barely matched baseline (Table 4.1). Diagnosing the training logs revealed three causes: black border padding from zoom-out (<1.0), double bilinear resampling on pre-resized images, and 30-epoch under-convergence.

Table 4.1: Phase 1 CPU baseline results (30 epochs, constant lr = 0.001, 3-seed validation means).

Phase 1 CPU Variant	Val Dice	Val IoU	Test Dice	Test IoU	Observed Limitation
`none` (Baseline, 3 seeds)	0.8084	0.6784	0.8471 (s42)	0.7348 (s42)	Overfits small training set
`flip` (p=0.5, 3 seeds)	0.8253	0.7027	0.8573 (s42)	0.7503 (s42)	Consistent gain over baseline
`both` (0.9–1.1×, p=1.0)	0.8096	0.6801	0.8433 (3-seed)	0.7292 (3-seed)	Black zoom-out padding & double blur

After applying zoom-in scaling (1.0–1.15×, $p = 0.5$), single-step raw stretch resizing, and 50-epoch cosine annealing on GPU, every augmented variant outperformed the baseline on both validation and held-out test splits (Table 4.2). Paired per-image permutation tests across the 26 held-out test images (Table 4.3) confirm that both `both` ($p = 0.00130$) and `both + copypaste` ($p = 0.00049$) achieve statistically significant IoU improvements over `none`.

Table 4.2: Master 6-way tuned GPU results across seeds 42, 43, and 44 (Mean ± Sample SD).

Variant	Val P	Val R	Val Dice / F1	Val IoU	Test P	Test R	Test Dice / F1	Test IoU
`none` (Baseline)	0.8323	0.7849	0.8078 ± 0.0062	0.6776 ± 0.0087	0.8364	0.8632	0.8495 ± 0.0130	0.7385 ± 0.0197
`zoom` only	0.8339	0.8006	0.8168 ± 0.0061	0.6903 ± 0.0088	0.8270	0.8806	0.8529 ± 0.0054	0.7436 ± 0.0082
`flip` only	0.8189	0.8372	0.8277 ± 0.0027	0.7061 ± 0.0039	0.8215	0.9105	0.8629 ± 0.0181	0.7592 ± 0.0280
`both` (Core Proposed)	0.8158	0.8448	0.8298 ± 0.0087	0.7091 ± 0.0127	0.8266	0.9113	0.8668 ± 0.0016	0.7649 ± 0.0025
`copypaste` only	0.8304	0.8691	0.8491 ± 0.0115	0.7379 ± 0.0174	0.8182	0.9130	0.8628 ± 0.0036	0.7588 ± 0.0056
`both + copypaste`	**0.8344**	**0.8696**	**0.8514 ± 0.0091**	**0.7414 ± 0.0138**	**0.8355**	**0.9132**	**0.8726 ± 0.0045**	**0.7740 ± 0.0070**

Table 4.3: Paired per-image statistical significance tests on held-out test images (n = 26, 3-seed mean per image).

Paired Comparison	Δ Dice	*t* (Dice)	*p* (Dice)	Δ IoU	*t* (IoU)	*p* (IoU)	Test Wins
`none → both`	+0.0164	3.26	0.00256	+0.0255	3.51	0.00130	20 / 26
`none → both + copypaste`	**+0.0233**	**3.38**	**0.00137**	**+0.0361**	**3.76**	**0.00049**	**22 / 26**
`both → both + copypaste`	+0.0069	1.39	0.18266	+0.0106	1.49	0.15218	15 / 26

4.3 Comparison with Existing Models and Ablation Analysis

Isolating each augmentation component clarifies their complementary roles: **horizontal flip** drives the largest single geometric gain (+2.07 pp test IoU over baseline) by doubling left-right pedestrian orientations, but exhibits high seed variance (±0.0280 IoU SD). **Small zoom** alone adds modest mean accuracy (+0.51 pp test IoU), yet when combined with flip (`both`) it acts as a strong regularizer that slashes test IoU seed variance nearly eight-fold (±0.0197 → ±0.0025). This +2.64 pp test IoU gain aligns directly with published semantic segmentation benchmarks (e.g., Romera et al. [7], reporting +1.0 to +2.5 pp IoU from geometric flip/scale jitter). Finally, adding **Copy-Paste** resolves the slight Precision dip observed in `flip` / `both` (recovering Precision from 0.8158 to 0.8344 on validation and 0.8266 to 0.8355 on test) while pushing Recall above 0.913, consistent with Ghiasi et al. [6].

4.4 Graphical Analysis

Figures 4.1–4.5 visualize the 6-way metric progression, 50-epoch convergence curves, seed-to-seed variance, per-image test IoU changes, and qualitative test-mask overlays.

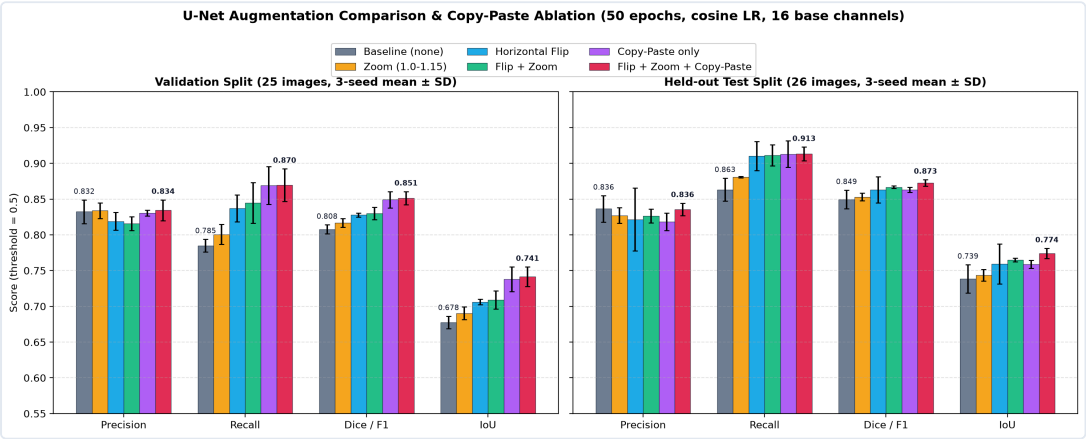


Figure 4.1: Grouped bar chart comparing 3-seed mean Precision, Recall, Dice/F1, and IoU across all six augmentation variants.

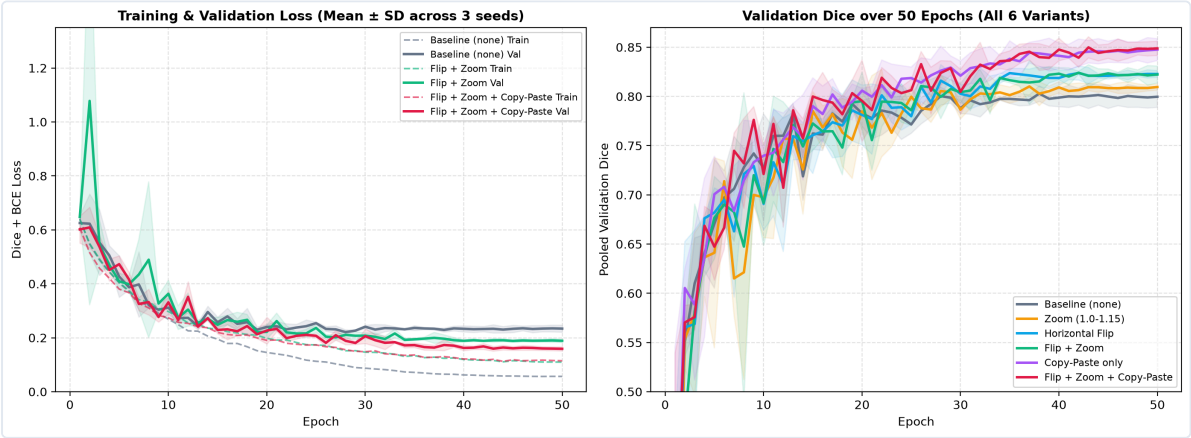


Figure 4.2: 50-epoch training loss, validation loss, and validation Dice trajectories. Augmentation narrows the train–validation generalization gap.

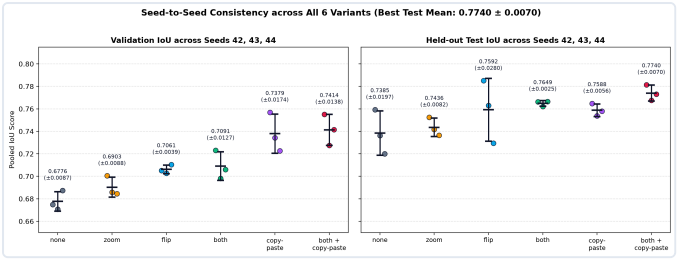


Figure 4.3: Seed-to-seed IoU consistency across seeds 42, 43, and 44 on validation and test splits.

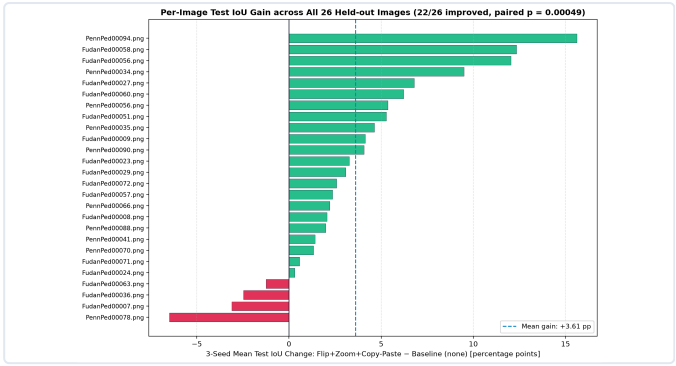


Figure 4.4: Per-image held-out test IoU change (ΔIoU) comparing `both + copypaste` vs. `none`.

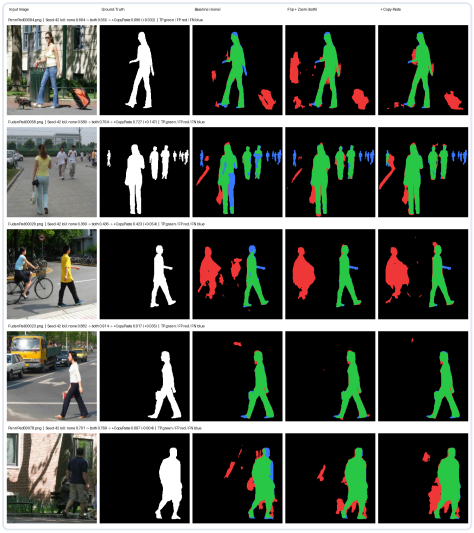


Figure 4.5: Qualitative mask comparison across five held-out test images (`FudanPed00058` , `FudanPed00023` , `FudanPed00036` , `PennPed00094` , `FudanPed00029`).

4.5 Discussion of Findings

Inspecting all 26 held-out test overlays for `tuned_both_copypaste_seed42` (micro Dice 0.8773, micro IoU 0.7815) explains both where augmentation succeeds and why residual errors persist (Table 4.4):

Table 4.4: Qualitative error-analysis breakdown on representative held-out test images (`tuned_both_copypaste_seed42`).

Test Image	Dice	IoU	Observed Behavior & Root Cause
`FudanPed00072.png`	0.961	0.925	Crisp pedestrian silhouette; residual error limited to 1–2 px boundary edges.
`FudanPed00023.png`	0.957	0.917	Clean foreground separation with minimal background noise.
`FudanPed00056.png`	0.903	0.823	Copy-Paste lifts Dice from 0.778 (`both`) and cuts FPs from 3,656 px to 446 px.
`PennPed00094.png`	0.821	0.696	Eliminates tree/bin shadow and small leashed dog FPs (4,936 → 2,613 px); suitcase remains.
`FudanPed00058.png`	0.842	0.727	Recovers foreground woman's leg and mid-ground people vs. baseline (0.580 IoU); tiny edge figures remain hard.
`FudanPed00036.png`	0.707	0.546	Low-contrast white shirt merges into pale wall (1,619 FN); doorway reflection causes 2,232 FP.
`FudanPed00029.png`	0.595	0.423	Model correctly segments an **unannotated cyclist** (6,109 FP, 94.8% Recall) missing from Penn-Fudan GT.

CHAPTER 5

5. Conclusion and Future Scope

5.1 Key Findings

- **Controlled Augmentation Substantially Improves Small-Dataset U-Net Generalization:** Combining horizontal flip and zoom-in scaling (1.0–1.15×) with single-step raw resizing and 50-epoch cosine annealing improved held-out test IoU from 0.7385 ± 0.0197 to 0.7649 ± 0.0025 (+2.64 pp, p = 0.00130) while cutting seed-to-seed variance nearly eight-fold.
- **Copy-Paste Recovers Precision and Reaches Best Overall Accuracy:** Extracting 290 pedestrian cutouts strictly from the 119 training images pushed validation Dice to 0.8514 (+4.36 pp over baseline) and held-out test Dice/IoU to 0.8726 / 0.7740 (+2.31 / +3.55 pp, 22/26 test images improved, p = 0.00049).
- **Preprocessing and Optimization Hygiene Matter:** Zoom-out black border padding and double bilinear resampling can mask real augmentation gains unless eliminated.

5.2 Limitations

- **Fixed 256 × 256 Input Resolution:** Downsampling wide street scenes (up to 1017 px) compresses distant pedestrians below 15 px wide, limiting small-person recall.
- **No External Pretraining:** Training a 16-channel U-Net from scratch leaves the model vulnerable to camouflage-like low-contrast clothing/wall boundaries and glass reflections (e.g., `FudanPed00036.png`).
- **Single Dataset & Annotation Noise:** Evaluation rests on 26 Penn-Fudan test images, where at least one image (`FudanPed00029.png`) contains an unannotated cyclist that penalizes Precision.

5.3 Future Research Directions

- Evaluate higher input resolutions (e.g., `512 × 512`) or ImageNet-pretrained encoders (ResNet34 / MobileNetV3 U-Net) under the exact same leak-free split.
- Add photometric jitter (brightness, contrast, slight color shifts) alongside Copy-Paste to reduce sensitivity to doorway reflections and harsh shadows.
- Test cross-dataset generalization on larger pedestrian benchmarks such as CityPersons or EuroCity Persons.

END MATTER

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Appendix: Repository Structure & Reproducibility

All code, fixed splits, 18 tuned GPU model checkpoints, evaluation JSONs, and consolidated deliverables are versioned in `person_segmentation_unet` (Table A.1).

Table A.1: Repository module inventory and reproducible CLI commands.

Module / Directory	Purpose
`src/person_segmentation/analyze_dataset.py`	Audits 170 pairs and generates `report/dataset_analysis/split_seed42.json`.
`src/person_segmentation/data.py` & `model.py`	Single-step stretch/letterbox dataset loader, augmentations, and 16-ch U-Net.
`src/person_segmentation/train.py`	Trains `none`, `zoom`, `flip`, `both` with cosine LR & RAM caching.
`src/person_segmentation/copypaste.py` & `train_copypaste.py`	Extracts 290 training-only cutouts and trains Copy-Paste variants.
`src/person_segmentation/evaluate.py` & `plot_results.py`	Computes pooled/per-image metrics and generates all 5 publication figures.
`results/all_project_results/`	Consolidated archive of metrics CSV, checkpoints, masks, and reports.

Example GPU Training Command: `python -m person_segmentation.train --aug-mode both --seed 42 --epochs 50 --lr-scheduler cosine --cache --output-dir results/tuned_both_seed42`